

Metric-Aware Test-Time Adaptation for Cross-Subject Multimodal Epileptiform-Discharge Detection

Yuting Yan
Central South University
FuRong Laboratory
Changsha, China
University of Science and Technology
of China
Institute of Advanced Technology
Hefei, China
ytyan@mail.ustc.edu.cn

Jiaqi Zhang
Jiangsu University
School of Computer Science and
Telecommunication Engineering
Zhenjiang, China
3230602065@stmail.uj.edu.cn

Ziliang Wang*
Central South University
School of Artificial Intelligence
Changsha, China
FuRong Laboratory
Changsha, Hunan, China
wangziliang@csu.edu.cn

Abstract

Cross-subject multimodal epileptiform-discharge detection is of significant clinical value. Most existing methods focus only on improving the training-stage backbone or the multimodal fusion strategy; however, the detection patterns learned from the source patient population become unreliable when transferred to an entirely new patient cohort. We propose a test-time adaptation framework that builds on the structure of the unlabeled target cohort. For the binary spike classification task, the model must reliably distinguish the relative risk of candidate segments. We exploit EEG-video neighbourhood consistency within the target cohort and apply a single Joint Multimodal Transductive Re-ranking (JMTR) step to the spike scores of the frozen detector, adapting the detection scores under cross-subject distribution shift. For the 5-class seizure subtype classification task, the model must produce stable class decisions, yet subtype priors differ substantially across patients. We estimate the drift between the predicted class distribution and the training prior, and construct a Drift-Gated Calibration (DGC) that gates between training-prior anchoring and distribution-preserving smoothing to correct the class posteriors, keeping the model robust under cross-subject prior shift. On the two NeuroMM-2026 tasks, we validate the effectiveness of the test-time adaptation framework.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Health informatics*; • **Information systems** → *Multimedia information systems*.

Keywords

EEG interictal epileptiform discharge detection, multimodal fusion, cross-subject generalization, test-time adaptation, distribution shift

ACM Reference Format:

Yuting Yan, Jiaqi Zhang, and Ziliang Wang. 2026. Metric-Aware Test-Time Adaptation for Cross-Subject Multimodal Epileptiform-Discharge Detection.

*Corresponding author.



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ACM ISBN 979-8-4007-2213-4/2026/11
<https://doi.org/10.1145/3767308.3837696>

In *Proceedings of the 34th ACM International Conference on Multimedia (MM '26)*, November 10–14, 2026, Rio de Janeiro, Brazil. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3767308.3837696>

1 Introduction

Interictal epileptiform discharges (IEDs) are the electrographic signatures that guide epilepsy diagnosis, and detecting them in *patients the model has never seen* is the clinically realistic regime: a deployed system is always applied to a new person. This makes IED analysis inherently cross-subject and multimodal. Each recording is multi-physiological—a 23-channel scalp-EEG montage (10–20 system) together with electrocardiogram (ECG) and electromyogram (EMG) channels—and is paired with synchronized video that captures behavioural correlates of a discharge. A discharge is thus observed simultaneously through electrical, cardiac, muscular, and visual channels, and a useful detector must fuse them while remaining robust to the fact that *the next patient is not the previous one*.

Existing methods for epileptiform-discharge analysis mostly focus on improving the training-stage backbone or multimodal fusion strategy, including compact EEG networks and self-supervised EEG representations [10, 12, 13, 21, 24], as well as reliability-aware multimodal fusion [2, 5, 6]. However, a deployed detector is applied to new patient cohorts whose data statistics may differ substantially from those of the source patients, a well-documented difficulty in cross-subject EEG analysis [7, 9, 18, 26]. Class priors, signal characteristics, and modality reliability learned during training can therefore become unreliable at test time, reflecting broader dataset-shift effects in both label and input distributions [19, 23]. This creates a mismatch between the predictions of a strong frozen model and the structure of the incoming target cohort.

The unlabeled target cohort itself contains information that can be used for adaptation, but what should be adapted at test time depends on the output requirement of the task. In epileptiform-discharge detection and screening, the system must reliably distinguish the relative risk of candidate segments: high-risk segments should be ranked ahead of low-risk ones. In seizure subtype classification, the system must instead make stable class decisions under patient-dependent changes in subtype prevalence. These two settings require different forms of test-time correction: the former should adjust detection scores according to the structure of the target cohort, whereas the latter should adapt class posteriors while avoiding misleading prior correction under prior shift.

We therefore propose a metric-aware test-time adaptation framework for frozen multimodal detectors. The framework uses only the unlabeled target cohort, does not update network weights, and matches the adaptation target to the output requirement of each task. For detection and screening, evaluation depends on the relative risk order of candidate segments, while order-preserving calibration cannot change this order. We therefore instantiate this branch as Joint Multimodal Transductive Re-ranking (JMTR), which exploits EEG-video neighbourhood consistency to perform one anchored re-ranking step on the frozen detector’s spike scores. For seizure subtype classification, the model must make stable class decisions, but patient-dependent subtype-prior shift makes blind anchoring to the training prior risky. We therefore instantiate this branch as Drift-Gated Calibration (DGC), which measures the distance between the target cohort’s predicted class distribution and the training prior using an unlabeled drift statistic, and gates between training-prior anchoring and distribution-preserving neighbourhood smoothing. Both branches follow the same principle: adapt only the part of the output required by the task, and trust only the evidence supported by the unlabeled target cohort. Figure 1 gives an overview of the framework and its two task-matched branches.

We validate the framework on NeuroMM-2026 [15, 16], a cross-subject benchmark of 84 patients with patient-disjoint training and test splits. The benchmark contains two complementary tasks aligned with the above deployment requirements: binary spike detection from EEG and video, evaluated by AUPRC, and five-class discharge-subtype recognition from EEG, evaluated by weighted-F1. Our framework improves the frozen detection backbone and provides drift-robust subtype calibration.

Our contributions are summarized as follows:

- We propose a metric-aware, label-free test-time adaptation framework that keeps the backbone frozen and adapts only the prediction object required by each task.
- We instantiate the framework with two task-matched branches. JMTR uses joint EEG-video neighbourhood consistency to perform an anchored re-ranking step for detection/screening, while DGC uses an unlabeled cohort-drift statistic to gate between training-prior anchoring and distribution-preserving smoothing for subtype classification.
- Experiments and ablations on NeuroMM-2026 show gains from matching the adaptation target to the task output: score order for detection and class posteriors for subtype recognition.

2 Related Work

EEG-based epileptiform-discharge detection. Modern epileptiform-discharge detectors are predominantly deep models over raw EEG: compact convolutional networks [13, 24] and self-supervised foundation models such as BENDR [12], LaBraM [10], and REVE [21] pre-trained on large corpora [20, 27].

Pairing EEG with synchronized video further improves detection sensitivity for multimodal IED benchmarks [14, 15]. These methods target the *backbone* and are typically evaluated pooled or within-subject.

Cross-subject generalization to unseen patients is a recognized distribution-shift problem, classically attacked by aligning subject

distributions [7, 9]; what a backbone should adapt *at test time*, and under which metric, remains open. Our contribution is backbone-agnostic: we operate post hoc and label-free on a frozen model’s output probabilities, and ask which adaptation the metric admits.

Multimodal fusion and reliability. A large literature fuses modalities by estimating how much to trust each one. Trusted multi-view classification combines modalities through evidential, Dempster-Shafer opinion aggregation weighted by per-view uncertainty [6], and dynamic multimodal fusion reweights modalities per sample by their informativeness [5]. The common ingredient is *per-modality reliability*. Our point of departure is that this reliability is estimated on the source distribution and need not transfer to a new patient, so reliability-weighted fusion degrades under cross-subject shift. JMTR uses the modalities through structure: it forms a joint EEG+video neighbourhood over the test cohort and re-ranks each segment by its multimodal neighbours, with no per-modality reliability weights and no retraining.

Test-time adaptation, label shift, and cross-subject generalization. Test-time adaptation adjusts a frozen model to an unlabeled target stream: entropy minimization and feature-statistic updates adapt the network [29], while label-shift methods recalibrate outputs to a target class prior [1, 4, 17, 25]. These assume the target prior is estimable, but that assumption fails under large shift [4], and post-hoc calibration degrades under drift [22, 28]. DGC gates calibration by a label-free drift statistic and otherwise preserves the cohort’s own distribution. Such calibration is rank-invariant and cannot affect ranking metrics, which determines where it may be used.

3 Method

3.1 Problem Formulation

We consider label-free test-time adaptation for cross-subject epileptiform-discharge analysis. Let $\mathcal{D}_s = \{(x_i^s, y_i^s)\}$ be a labeled source cohort used to train a backbone f_θ , and let $\mathcal{D}_t = \{x_i^t\}_{i=1}^N$ be an unlabeled target cohort collected from patients unseen during training. At test time, f_θ is frozen. The adaptation procedure may use the target inputs $\mathcal{D}_t$, target-side feature representations, and the predictions of f_θ on $\mathcal{D}_t$, but it cannot use target labels or update θ . The goal is to transform the frozen predictions into adapted predictions that better match the incoming patient cohort.

Clinical epileptiform-discharge analysis naturally contains two types of prediction problems with different output objects.

Detection and screening. Given a candidate segment, a frozen detector outputs a score

$$s_i = f_\theta(x_i^t) \in [0, 1],$$

forming a score vector $\mathbf{s} = (s_1, \dots, s_N)$ over the target cohort. For screening, the operational object is the induced rank order

$$\sigma_{\mathbf{s}} : s_{\sigma(1)} \geq s_{\sigma(2)} \geq \dots \geq s_{\sigma(N)},$$

because high-risk segments must be reviewed first. Any correction that preserves $\sigma_{\mathbf{s}}$ cannot change the screening order; test-time adaptation must therefore change the ordering itself.

Seizure subtype classification. A frozen classifier outputs a posterior vector $\mathbf{p}_i \in \Delta^K$, and the target-cohort output is a posterior

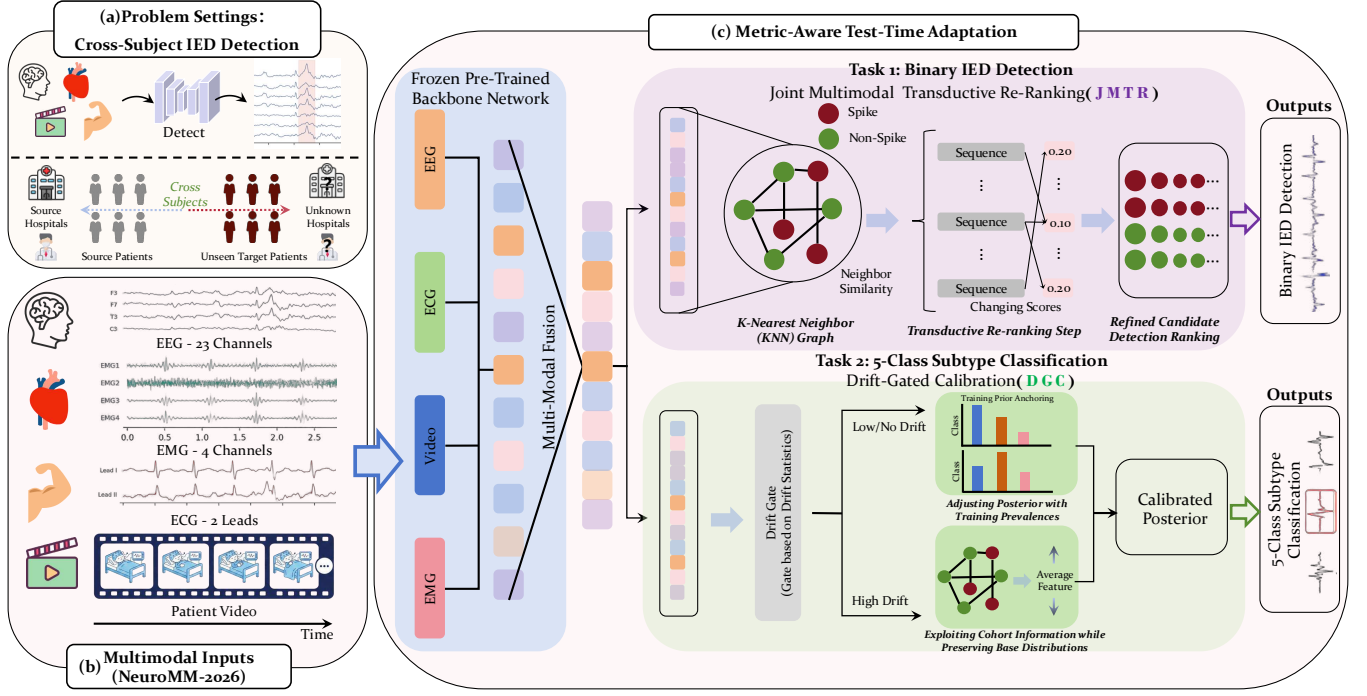


Figure 1: Overview of the proposed metric-aware test-time adaptation framework. The framework adapts frozen multimodal predictions using unlabeled target-cohort evidence, with separate correction targets for detection/screening (JMTR) and seizure subtype classification (DGC).

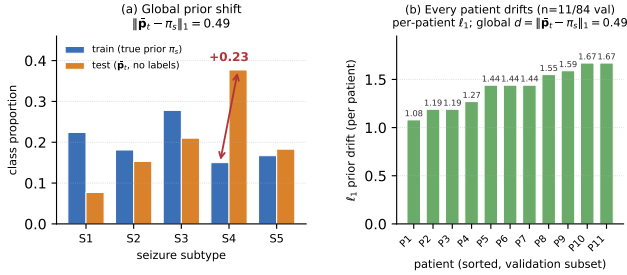


Figure 2: Cross-subject prior shift (validation; test is label-free). (a) Global prior shift ($\|\bar{\mathbf{p}}_t - \pi_s\|_1 = 0.49$). (b) Per-patient ℓ_1 drift (11 validation patients, subset of 84); every patient drifts substantially.

matrix $P = [\mathbf{p}_1; \dots; \mathbf{p}_N]$, with decisions $\hat{y}_i = \arg \max_k p_{ik}$. Under cross-subject deployment, subtype prevalence varies. Let

$$\pi_s = \frac{1}{|\mathcal{D}_s|} \sum_{(x,y) \in \mathcal{D}_s} \text{onehot}(y), \quad \bar{\mathbf{p}}_t = \frac{1}{N} \sum_{i=1}^N \mathbf{p}_i.$$

A large mismatch between $\bar{\mathbf{p}}_t$ and π_s indicates that blindly reusing source-side priors may be unreliable (Figure 2).

Thus, epileptiform-discharge analysis exposes two different adaptation targets: detection and screening require adapting the relative ordering of scores $\mathbf{s}$, whereas subtype classification requires adapting the posterior matrix P , and these two targets call for different correction mechanisms.

3.2 Metric-Aware Test-Time Adaptation

The formulation above suggests that adaptation should not apply the same correction to every prediction object. A detection score vector and a subtype posterior matrix serve different clinical purposes: the former supports risk ordering, whereas the latter supports class decisions. We therefore make the adaptation target metric-aware.

For detection, the relevant object is the score ordering. Since calibration leaves AUPRC unchanged (Figure 3), the adaptation target must be a cohort-supported re-ranking. In our framework, this evidence comes from neighbourhood consistency in the target cohort: segments that are close in the multimodal feature space should support similar detection scores.

For subtype classification, the relevant object is the posterior matrix P and decisions $\arg \max_k p_{ik}$. The source prior should not be trusted blindly under cross-subject shift; adaptation strength should depend on cohort-source compatibility. In our framework, this compatibility is measured by an unlabeled drift statistic computed from the predicted target class distribution.

Both cases are implemented through the same bounded update:

$$\hat{O} = O_{\text{base}} + g(O_{\text{adapt}} - O_{\text{base}}), \quad g \in [0, 1], \quad (1)$$

where O_{base} is the frozen backbone output, O_{adapt} is a target-cohort-supported adaptation target, and g controls how far the prediction moves away from the backbone. This form keeps the adaptation conservative: when target-side evidence is weak or unreliable, g can shrink the update toward the frozen prediction; when the evidence is task-relevant, g allows the prediction to move toward the adapted

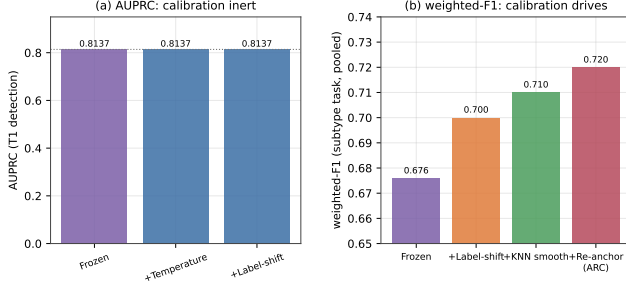


Figure 3: Calibration is rank-invariant (validation). The same calibration leaves AUPRC unchanged (a) but drives weighted-F1 (b); hence detection (JMTR) must re-rank while seizure subtype classification (DGC) calibrates. ARC = Anchored Re-ranking Correction (the JMTR step). Values are on the validation split.

target. The next two subsections instantiate this framework for the two prediction objects. For detection and screening, $O_{\text{base}} = s$ and O_{adapt} is obtained by joint EEG-video neighbourhood re-ranking (JMTR). For subtype classification, $O_{\text{base}} = P$ and O_{adapt} is selected by a drift gate between source-prior anchoring and distribution-preserving smoothing (DGC).

3.3 Task-Specific Branches: JMTR and DGC

The two branches are parallel instances of the bounded update (Eq. 1), one per (task, metric) pair: detection and screening, scored by AUPRC, is handled by JMTR; seizure subtype classification, scored by weighted-F1, is handled by DGC. They differ in the output object adapted—score vector s vs. posterior matrix P —and in how the gate is set: a fixed step for JMTR and a drift-dependent gate for DGC.

Detection and screening (JMTR). For detection and screening, the adaptation target is the relative ordering of spike scores. A frozen detector provides a score vector $s = (s_1, \dots, s_N)$ on the target cohort, but these scores may be affected by patient-specific signal characteristics and modality reliability. Since screening relies on which candidate segments are ranked earlier, a useful test-time correction should change the ordering only when the unlabeled target cohort provides structural evidence for doing so. We instantiate this branch as *Joint Multimodal Transductive Re-ranking* (JMTR). The key idea is to use the neighbourhood structure of the target cohort as label-free evidence. For each target segment, we construct an EEG feature $\phi_{\text{eeg}}(x_i)$ and a video feature $\phi_{\text{vid}}(x_i)$. The EEG feature is computed from per-channel relative band power over five bands ($\delta, \theta, \alpha, \beta, \gamma$) across the 29 recording channels (23 EEG + 6 ECG/EMG; 145 dimensions), and the video feature is a self-supervised vision-transformer (DINOv2) embedding (768 dimensions). The two feature blocks are standardized and scale-balanced, then concatenated into a joint representation

$$z_i = [\text{norm}(\phi_{\text{eeg}}(x_i)), \lambda \text{norm}(\phi_{\text{vid}}(x_i))],$$

where $\lambda = \sqrt{145/768}$ balances the two blocks by their dimensionality. We then build a k -nearest-neighbour graph ($k = 50$) over $\{z_i\}_{i=1}^N$. This joint graph treats two segments as neighbours only when they are similar under the combined EEG-video representation.

Implementation. We use exact nearest-neighbour search (SCIKIT-LEARN NearestNeighbors, brute-force Euclidean, $k=50$, complexity $O(N^2d)$, $d=913$). For $N \approx 20,000$ (NeuroMM) this takes ~ 1.4 s on a single CPU core with a ~ 1.5 GB distance matrix; smaller cohorts are cheaper ($N=1,000$: 0.18 s, 4 MB; $N=5,000$: 0.17 s, 95 MB). Approximate methods (e.g. FAISS) would reduce cost further for very large cohorts but were not needed here. Hyperparameters were fixed on OOF validation; see §5.2 for stability.

Given the neighbourhood $\mathcal{N}_k(i)$ of segment i , JMTR defines a cohort-supported score target by averaging the frozen scores of its neighbours:

$$s_i^{\text{adapt}} = \frac{1}{k} \sum_{j \in \mathcal{N}_k(i)} s_j.$$

The final adapted score follows the bounded update in Eq. 1:

$$\hat{s}_i = (1 - \alpha)s_i + \alpha s_i^{\text{adapt}}, \quad \alpha = 0.4.$$

We use a single anchored step rather than iterating propagation to convergence. This keeps the correction close to the frozen detector while allowing local multimodal evidence to adjust the risk order of candidate segments. The procedure uses no target labels and does not update the detector; it only reorders spike scores using unlabeled target-cohort structure.

Seizure subtype classification (DGC). For subtype classification, a frozen classifier outputs posteriors $p_i \in \Delta^K$ with decisions $\hat{y}_i = \arg \max_k p_{ik}$. Calibrating toward the source prior can change these decisions, but under cross-subject shift the prevalence may differ substantially, so anchoring to the source prior helps only when the cohort still resembles training. A useful correction should therefore adapt posteriors by an amount that depends on how compatible the cohort is with the source prior. We instantiate this branch as *Drift-Gated Calibration* (DGC), which gates between two posterior operators by a label-free drift statistic. We first measure the discrepancy between the predicted target distribution and the source prior,

$$\bar{p}_t = \frac{1}{N} \sum_{i=1}^N p_i, \quad d = \|\bar{p}_t - \pi_s\|_1,$$

computed on the raw, un-anchored posteriors so that it reflects the true cohort distance to training. The drift is mapped to a gate

$$g = \text{clip}\left(\frac{d_{\text{hi}} - d}{d_{\text{hi}} - d_{\text{lo}}}, 0, 1\right), \quad d_{\text{lo}} = 0.20, d_{\text{hi}} = 0.40,$$

so that a small drift opens the gate ($g \rightarrow 1$) and a large drift closes it ($g \rightarrow 0$). DGC combines two operators on the posteriors. The anchoring operator $A(\cdot)$ aligns the cohort class marginal to π_s by iterative proportional fitting [1]; the smoothing operator $S(\cdot)$ replaces each posterior by a k -nearest-neighbour average in log-probability space ($k = 200$), which denoises the predictions while preserving the cohort's own class marginal.

Taking the distribution-preserving smoothing as the conservative base and the prior anchoring as the adaptation target, the final posterior follows the same bounded update as Eq. 1:

$$\hat{p}_i = S(p_i) + g(A(p_i) - S(p_i)).$$

When the cohort matches the source prior the gate opens and recovers the calibration gain; when the cohort has drifted it closes and falls back to distribution-preserving smoothing. Like JMTR,

DGC uses no target labels and does not update the classifier; it adapts only the posteriors using the unlabeled target cohort.

4 Experiments

4.1 Experimental Setup

Dataset. We evaluate on NeuroMM-2026, a cross-subject benchmark for multimodal epileptiform-discharge analysis. The dataset contains recordings from 84 patients, and the training and test cohorts are patient-disjoint. The benchmark covers two complementary clinical tasks. The first is epileptiform-discharge detection and screening from synchronized EEG and video. The second is seizure subtype classification from EEG, where each discharge candidate is assigned to one of five subtypes. In both tasks, the target cohort is available as an unlabeled batch at test time, while target labels are not used by our adaptation methods.

To probe the subtype branch under a *different* drift regime, we additionally evaluate DGC on Sleep-EDF [11], a public polysomnography corpus whose five-class sleep-stage labels give a comparable five-way decision task. Unlike the strongly shifted NeuroMM cohort, Sleep-EDF is a *matched* cohort: its class marginal stays close to the source prior (small drift $d \approx 0.13$), so it serves as a control where anchoring to the prior *should* help.

Metrics. The detection task is evaluated by AUPRC. Given labels $y_i \in \{0, 1\}$ and scores s_i , at threshold τ :

$$\text{Prec}(\tau) = \frac{\sum_i \mathbf{1}[s_i \geq \tau] y_i}{\sum_i \mathbf{1}[s_i \geq \tau] + \epsilon}, \quad \text{Rec}(\tau) = \frac{\sum_i \mathbf{1}[s_i \geq \tau] y_i}{\sum_i y_i + \epsilon}.$$

AUPRC is the area under this curve; it evaluates the ordering of candidate segments and is therefore sensitive to re-ranking. The subtype task is evaluated by weighted-F1:

$$\text{F1}_k = \frac{2\text{TP}_k}{2\text{TP}_k + \text{FP}_k + \text{FN}_k + \epsilon}, \quad \text{wF1} = \sum_{k=1}^K \frac{n_k}{\sum_c n_c} \text{F1}_k.$$

Weighted-F1 evaluates the final decisions $\hat{y}_i = \arg \max_k p_{ik}$ and is therefore affected by posterior adaptation.

Evaluation protocol. All test-time adaptation operators are applied post hoc to frozen backbone outputs. During adaptation, we use only the unlabeled target cohort, target-side features, and the backbone predictions. We do not use target labels and do not update network weights. This protocol isolates the effect of the proposed test-time adaptation framework from improvements due to backbone training. All hyperparameters (α , the neighbourhood sizes k , and the drift thresholds d_{lo} , d_{hi}) are fixed on the labeled validation split, never on the held-out test set, so the procedure stays label-free.

4.2 Frozen Backbones

Our goal is not to introduce a new backbone architecture, but to adapt the outputs of strong frozen models to unseen patient cohorts. For detection, we use an early-fusion EEG-video detector that combines multi-band time-frequency EEG encodings with self-supervised video features and averages patient-disjoint fold predictions to obtain spike scores. For subtype classification, we

Table 1: Detection/screening: results and ablations (AUPRC, NeuroMM test set). All operators are post-hoc and label-free on the same frozen detector output.

Method	Adapts / evidence	AUPRC
Backbone (frozen)	–	0.9781
Calibration	scale	0.9781
LAME [3]	transductive	0.9577
LATA [32]	transductive	0.9804
T3A [8]	prototype-TTA	0.9432
Label propagation [31]	transductive	0.9865
Score alignment	per-cluster	0.9601
Agreement gate	EEG+video	0.9855
Smoothing [†]	video	0.9889
Smoothing [†]	EEG	0.9896
Smoothing [†]	EEG+video	0.9902
JMTR (ours)	EEG+video re-rank	0.9937

[†]Smoothing uses $\alpha=1$ (full neighbourhood average, no anchoring).

Table 2: Seizure subtype classification: results and ablations (weighted-F1). NeuroMM is the cross-subject test cohort; Sleep-EDF is an external matched cohort.

Method	Adapts / evidence	NeuroMM	Sleep-EDF
Backbone (frozen)	–	0.7772	0.7850
Label propagation [31]	posterior	0.7772	0.7848
LAME [3]	posterior	0.7817	0.7852
LATA [32]	posterior	0.7891	0.7852
T3A [8]	prototype-TTA	0.7621	0.7868
Prior anchoring	source prior	0.7581	0.7943
Smoothing [‡]	preserve dist.	0.7891	0.7852
DGC (ours)	drift-gated posterior	0.7891	0.7943

[‡]Smoothing: log-prob kNN average ($k=200$); distinct from Table 1 smoothing.

ensemble diverse EEG classifiers and combine their five-class posteriors by log-ratio geometric averaging followed by normalization. All adapters start from the same released frozen outputs.

5 Results

5.1 Main Results

Tables 1 and 2 give the overall comparison. The main observation is that the best correction is the one that matches the task output. For detection and screening, the task is driven by the relative order of candidate segments. Generic corrections that do not target this order help little (Table 1). Order-preserving calibration is rank-invariant and therefore leaves the detection ranking unchanged. LAME changes predictions, but it is not anchored to the strong detector score order and degrades performance in this low-positive-rate screening setting. In contrast, JMTR adapts the score order using EEG-video neighbourhood evidence and reaches 0.9937 AUPRC, improving the frozen detector by +0.0156.

For seizure subtype classification, the task is driven by stable per-sample class decisions. Generic posterior propagation barely helps (Table 2). DGC adapts class posteriors under cohort drift and raises the frozen classifier from 0.7772 to 0.7891 weighted-F1 on NeuroMM—the same level reached by LATA and Smoothing (Figure 4)—and from 0.7850 to 0.7943 on Sleep-EDF.

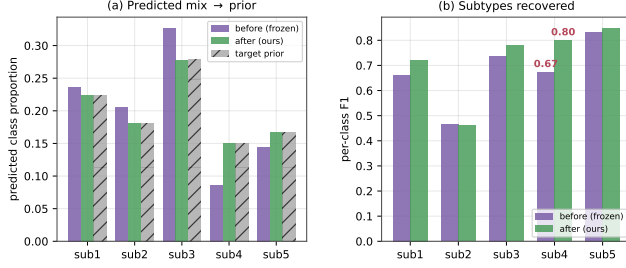


Figure 4: DGC recovers under-predicted subtypes (validation). Aligning the predicted class mix to the prior (a) raises per-class F1 (b; e.g. sub4 0.67 → 0.80).

Comparison with LATA and T3A. For detection, LATA [32] reaches 0.9804 AUPRC and T3A [8] reaches 0.9432 AUPRC; JMTR outperforms both by exploiting multimodal neighbourhood structure. GCR [30] (graph-convolutional re-ranking) was designed for closed-set person re-identification and related retrieval settings with a query-gallery similarity structure, while our setting is label-free cohort-level clinical screening without a query-gallery protocol. We therefore provide a conceptual comparison only. GCR propagates affinities over a graph of retrieval candidates, whereas JMTR performs a single closed-form anchored aggregation of frozen spike scores on a joint EEG-video kNN graph. For subtype classification, under the NeuroMM shift ($d=0.49 > d_{hi}=0.40$, $g=0$) DGC defaults to distribution-preserving smoothing and matches LATA and Smoothing (all 0.7891), yielding neither improvement nor degradation over these baselines. The advantage of DGC appears on the matched Sleep-EDF cohort ($d \approx 0.13$, $g=1$) where anchoring is appropriate: DGC attains 0.7943, recovering the calibration gain that the smoothing-based baselines forfeit (0.7852 for both LATA and Smoothing).

5.2 Ablation Study

We next ablate the two branches of the framework to identify where the gains come from. The ablation rows in Tables 1 and 2 report task-specific variants for detection/screening and seizure subtype classification.

For detection and screening, score alignment decreases AUPRC relative to the frozen backbone, and agreement gating is outperformed by neighbourhood smoothing alone. In contrast, neighbourhood smoothing consistently improves the frozen detector, and the joint EEG-video neighbourhood outperforms either single-modality neighbourhood. This confirms that multimodal neighbourhood consistency provides more informative test-time evidence than any single modality. The final anchored re-ranking branch obtains the best AUPRC.

For seizure subtype classification, the two fixed arms behave oppositely across cohorts: prior anchoring helps on matched Sleep-EDF but drops on shifted NeuroMM, while smoothing is safer on NeuroMM but loses calibration gain on Sleep-EDF. The drift gate matches the best arm on both.

Sensitivity. Defaults ($k=50$, $\alpha=0.4$, $d_{lo}=0.20$, $d_{hi}=0.40$) were chosen by grid search on held-out OOF validation maximising AUPRC for JMTR and weighted-F1 for DGC; no target labels were used. For

Table 3: Evidence motivating metric-aware test-time adaptation.

Quantity	Value	Takeaway
$\ \hat{p}_t - \pi_s\ _1$	0.49	Source prior unreliable
Median local mismatch	0.72	Pervasive local shift
Matched regions (< 0.20)	0%	Blind anchoring risky
Δ AUPRC (calibration)	0.0000	Re-ranking required

JMTR, varying $k \in [20, 200]$ and $\alpha \in [0.1, 0.6]$ gives AUPRC rank correlation $\rho \geq 0.997$ and $\rho \geq 0.988$ with the default; only $\alpha=0.8$ degrades ($\rho=0.936$). For DGC, $d=0.49 > d_{hi}=0.40$ on NeuroMM so $g=0$ for all threshold pairs tested—no per-dataset tuning needed; on Sleep-EDF $d \approx 0.13$ keeps $g=1$ across all pairs.

5.3 Evidence and Analysis

Table 3 summarizes why metric-aware adaptation is needed: the held-out cohort’s predicted class distribution is far from the source prior, this mismatch is pervasive down to the local-neighbourhood level, and order-preserving calibration leaves AUPRC unchanged while driving weighted-F1 (Figure 3). These observations explain the design: detection re-ranks spike scores (JMTR), while subtype classification uses drift-gated calibration (DGC). Notably, the zero matched regions (< 0.20) confirm that blind prior anchoring would be risky on this cohort, validating the conservative gating design of DGC. The median local mismatch of 0.72 further shows that the drift is not confined to the global class marginal but is pervasive at the segment level, underscoring why a cohort-wide drift statistic is sufficient to characterize the adaptation regime.

6 Conclusion

We proposed a label-free test-time adaptation framework for cross-subject multimodal epileptiform-discharge analysis that uses only the unlabeled target cohort and does not update network weights. The key insight is metric awareness: for detection and screening, the framework adapts spike-score order through anchored EEG-video neighbourhood re-ranking (JMTR); for seizure subtype classification, it adapts class posteriors through drift-gated calibration (DGC). Experiments on NeuroMM-2026 confirm that gains come from matching the adaptation target to the task output requirement, and drift statistics provide useful signals for cross-subject generalization. The principle generalises across tasks: let the evaluation metric determine what to adapt, and trust only unlabeled cohort structure. Both branches are post-hoc and label-free, making the framework directly applicable to any frozen multimodal backbone without retraining. The drift statistic computed from unlabeled predictions serves as a lightweight proxy for cohort compatibility, requiring no held-out labeled data and adding negligible overhead. We hope this work encourages future research to treat the evaluation metric as a first-class design constraint in clinical test-time adaptation.

Acknowledgments

This work was supported by the Scientific Research Program of FuRong Laboratory (No. 2025PT5009).

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